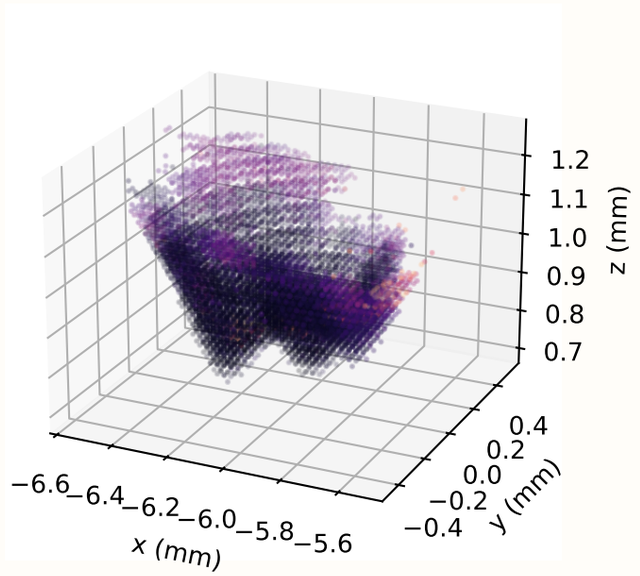
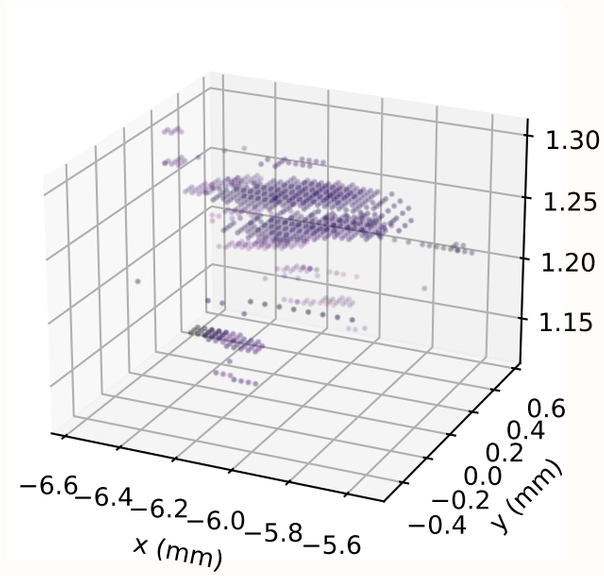
Native failure mechanism: local debris shedding with attached core

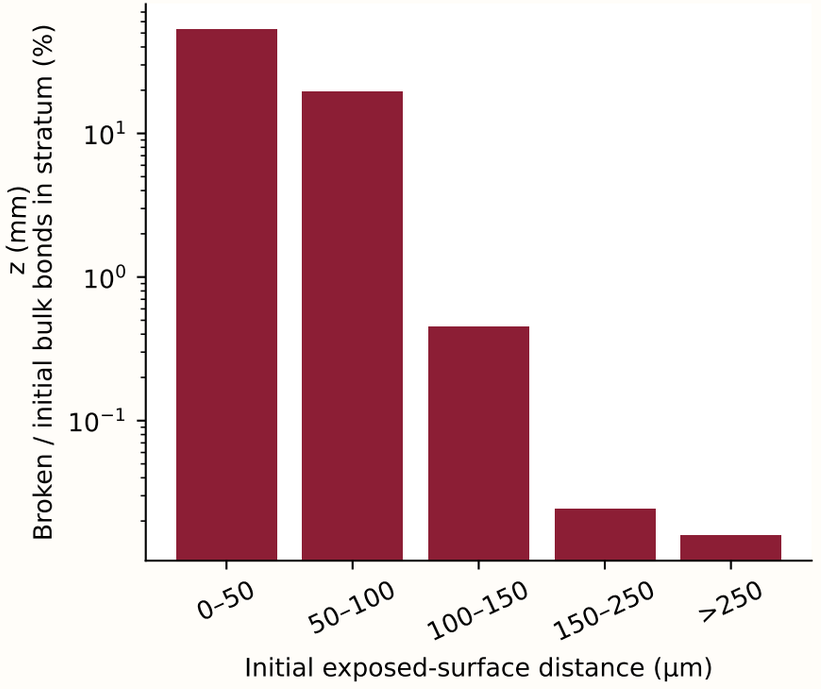
Bulk breaks at reference bond midpoints

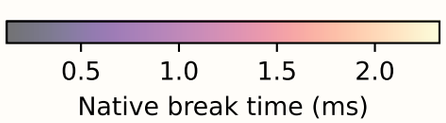


Interface breaks at reference bond midpoints

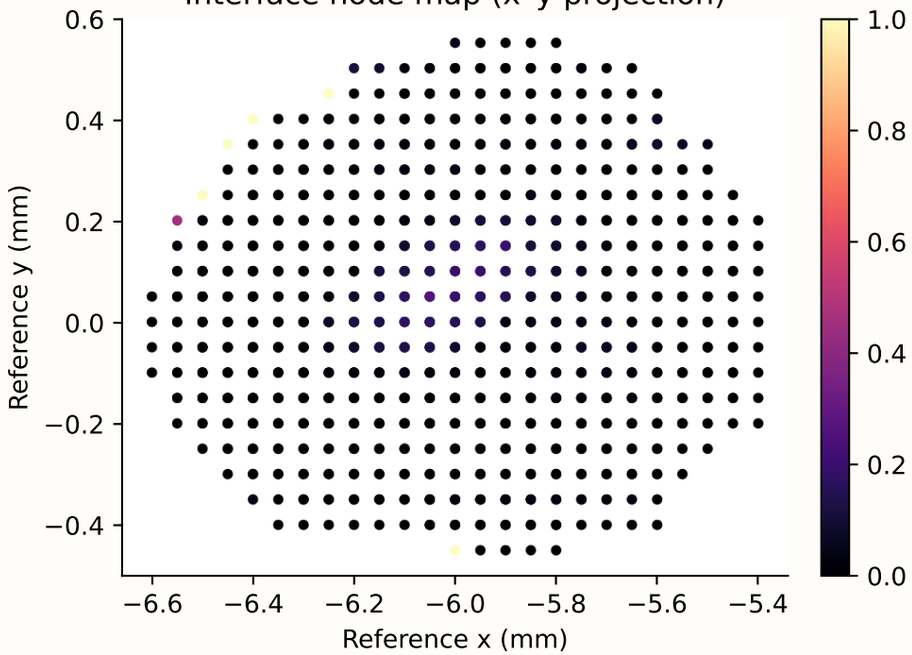


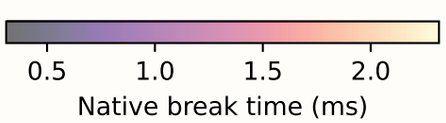
99.28% of broken bulk bonds lie within 100 µm



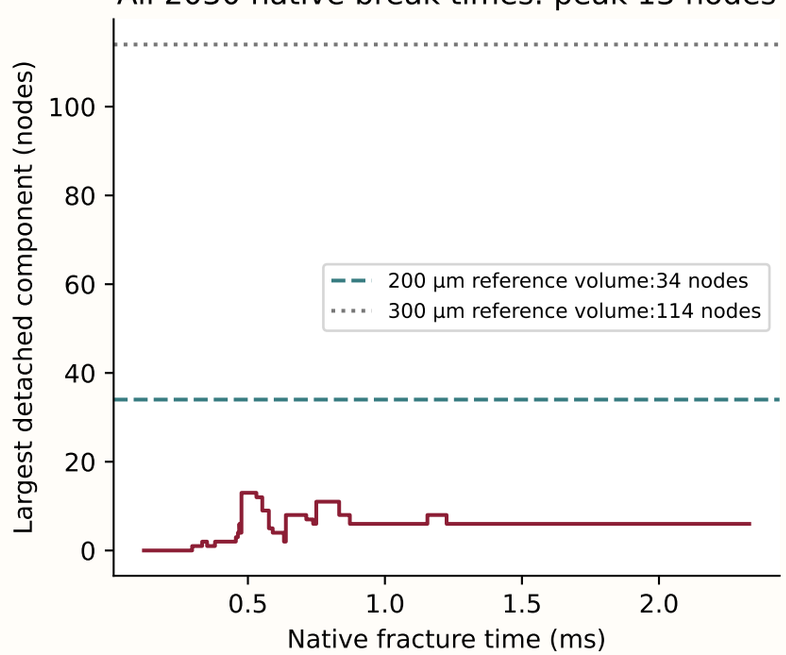


Interface node map (x–y projection)





All 2030 native break times: peak 13 nodes



Retained connectivity

1702 attached nodes:
1701 core +1 anchored singleton
70261 surviving core bulk bonds
21220 surviving core interface bonds
1282 distinct anchors connected to core

45 central axis sections tested
Minimum 184 surviving crossing bulk bonds

291 exact-member detached lifetimes
7 have no saved COM frame: explicitly unavailable

No established macroscopic peel front or cut surface.
Graph connectivity is evidence, not unique causation.